Assignment 2 (Software Requirements Specification - System Requirements)

Course code and name: ISYS3416/3432 Software Engineering Fundamentals for IT

Length: not more than 35 pages

Type: Written Report (with individual task)

Feedback mode: Written feedback via Canvas

Late work:

If you require an Extension of Submittable Work (assignments, reports project work, etc.) for 5 calendar days or less (from the original due date) and have valid reasons, you must complete and lodge an Application for an Extension of Submittable Work (5 Calendar Days or less) form and lodge it with the Senior Educator/ Program Manager. The application must be lodged no later than one working day before the official due date. You will be notified within no more than 2 working days of the date of lodgment as to whether the extension has been granted. If you seek an Extension of Submittable Work for more than 5 calendar days (from the original due date), you must lodge an Application for Special Consideration form under the provisions of the Special Consideration Policy, preferably before, but no later than 2 working days after the official due date. A submittable Work (assignments, projects, lab reports, etc.) that is submitted late without approval of an extension will not be accepted or marked.

Penalties for Late Submission: A penalty of 10% per day will apply to late assignments up to a maximum of five days (weekends count for two days, not one). Note that for assignments up to five days late due to illness, a medical certificate may be presented to your teacher and a late penalty will not apply.

Learning Objectives Assessed

- CLO2: Evaluate requirements for a software system
- CLO3: Apply the process of analysis and design using the object-oriented approach
- CLO4: Work effectively in a team to analyse the requirements of a complex software system and solve problems by creating appropriate designs that satisfy these requirements
- CLO5: Communicate effectively with others, especially regarding the progress of the system development and the contenct of the design, by means of reports and presentations. Use appropriate design, version control, and collaboration tools to work effectively as a team
- CLO7: Design and carry out tests using various testing techniques and tools

Ready for Life and Work

The following skills will be developed upon completing this course, such as:

- **Software Design Principles**—Understand abstraction, modularity, cohesion, coupling, and reusability.
- Be familiar with common design patterns (such as Singleton, MVC, etc.) and when to apply them.
- **UML Modeling**
  - Illustrate the system structure, attributes, operations, and relationships using class diagrams.
  - Shows the object interaction over time for specific scenarios using sequence diagrams.

Assessment Details

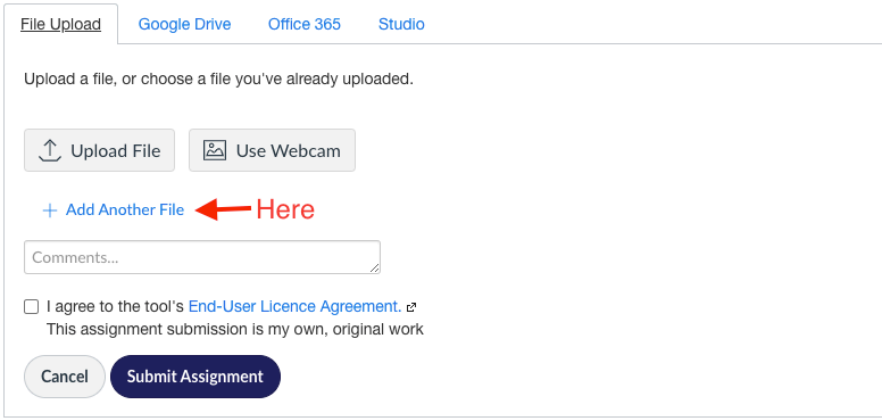
Based on the project topic assigned to you from Assessment 1, Assessment 2 (Milestone 2) will focus on the system requirements. These requirements specify how the software will meet the user's requirements, including both functional and non-functional aspects. Here are the requirements:

1. Create a user interface based on your design pattern for a specific problem scenario.
2. Draw UML diagrams (class and sequence).
3. Describe the software design pattern used.
4. **Each team member** will submit a peer evaluation form. Ensure that you provide comments on how each team member collaborates. Combine your peer evaluation grade and reflection into a single file. Here's the link to the submission: **Reflection & Peer Evaluation**

Use the template below to create a design document detailing the functionality of your systems. Download the template here: **A2 SRS Template** and the **peer evaluation template**

Submission Instruction

- Submit the **Group Report** to Turnitin together with the drawing **files** (***don't zip all files***). **Only one team member needs to submit this.**
  - Put all diagram drawing files (click the "**Add Another File**" option). If you use StarUML, please submit the *.mdj file. If you use Lucidchart, since it is online, it is fine to just submit the PDF/image version.
- Submit your **individual reflection and peer evaluation**



  - Name convention: *Names of* the *report files must follow the format "GroupNo_MaterialType". For example, Group8_FinalReport.docx, Group8_Diagrams.*

Support Resources

This assessment requires that you meet RMIT's expectations for academic integrity. More information and advice on how to avoid plagiarism are available in the Getting Started module.

- Open **the academic integrity page**.

Additional library and learning resources are available to help with the assessment in this course.

- Link to **Assignment Support**.

Rubric

Criteria	Ratings					Pts
Introduction Project Overview & Milestone Revision Explanation	**5 to >3.95 pts Excellent** Provides a clear, concise, and professional overview of the project scope and objectives. All revisions, additions, and removals from Milestone 1 are explicitly identified and logically categorized. Each change is clearly justified with strong reasoning. Demonstrate reflective thinking and strong alignment with project goals.	**3.95 to >3.45 pts Good** Provide a clear project overview. Most changes from Milestone 1 are identified and explained with reasonable justification. Minor gaps in depth of explanation or clarity. Overall alignment with project objectives is evident.	**3.45 to >2.95 pts Satisfactory** Provides a basic overview of the project. Some changes from Milestone 1 are mentioned, but explanations lack depth or clear justification. Limited reflection on why changes were necessary.	**2.95 to >2.45 pts Fair** Overview is brief but unclear or partially inaccurate. Changes are vaguely described or incompletely identified. Justifications are weak, unclear, or missing for several revisions.	**2.45 to >0 pts Poor** Project overview is missing, unclear, or inaccurate. No clear explanation of revisions, additions, or removals from Milestone 1. No justification provided.	5 pts
Design Patterns Application Creatively design and describe the design pattern applied in the application.	**10 to >7.93 pts Excellent** Demonstrates innovative and thoughtful selection of design pattern(s) that strongly enhance system architecture. The pattern is not only correctly applied but creatively adapted to the application's context. Clear, technically accurate explanation of structure, roles, interactions, and benefits	**7.93 to >6.93 pts Good** Appropriate design pattern selected and correctly applied. Clear explanation of how the pattern works and why it fits the system. Demonstrates good understanding of structural and behavioral impact. Minor gaps in depth of justification or creativity.	**6.93 to >5.93 pts Satisfactory** Design pattern correctly identified and generally applied, but explanation is descriptive rather than analytical. Limited creativity or adaptation to system context. Justification is basic and lacks deeper architectural reasoning.	**5.93 to >4.93 pts Fair** Pattern mentioned but weakly applied or not fully aligned with system needs. Explanation lacks clarity or contains minor conceptual errors. Minimal justification of architectural benefits. Limited evidence of creative thinking.	**4.93 to >0 pts Poor** Design pattern incorrectly selected, poorly applied, or missing. Explanation demonstrates misunderstanding of pattern structure or purpose. No meaningful justification or connection to system design.	10 pts

Criteria	Ratings					Pts
	(e.g. scalability, maintainability, flexibility). Strong justification with architectural reasoning and clear linkage to system requirements and class/sequence diagrams.					
UI/UX Design Creatively design UI/UX	**10 to >7.93 pts Excellent** UI/UX design demonstrates high creativity and originality while maintaining strong usability principles. Interfaces are intuitive, visually consistent, and aligned with user needs and use cases. Clear navigation flow, accessibility considerations, responsive design, and thoughtful interaction elements are evident. Design decisions are purposeful and enhance user experience significantly.	**7.93 to >6.93 pts Good** UI/UX design is visually appealing and mostly user-centered. Interfaces are logically structured with consistent layout and clear navigation. Demonstrates creativity with minor usability or consistency issues. Most user scenarios are effectively supported.	**6.93 to >5.93 pts Satisfactory** UI/UX design is functional and supports key tasks but lacks originality or refinement. Basic usability principles are applied, though some navigation, layout, or consistency issues exist. Limited evidence of creative enhancement.	**5.93 to >4.93 pts Fair** UI/UX design shows limited creativity and weak alignment with user needs. Navigation may be confusing, layouts inconsistent, or key screens missing. Usability principles are minimally applied.	**4.93 to >0 pts Poor** UI/UX design is incomplete, unclear, or not aligned with user requirements. Poor navigation, inconsistent layout, and lack of usability considerations. Minimal or no evidence of creative design thinking.	10 pts
UML Sequence Diagram Design and Description	**10 to >7.92 pts Excellent** All relevant use cases are fully represented with complete and logically	**7.92 to >6.92 pts Good** Most major use cases are represented with correct UML notation and	**6.92 to >5.92 pts Satisfactory** Several sequence diagrams are provided but may be incomplete or	**5.92 to >4.92 pts Fair** Few sequence diagrams provided or major use cases missing. UML	**4.92 to >0 pts Poor** Sequence diagrams are missing or fundamentally incorrect. Major	10 pts

Criteria	Ratings					Pts
	structured sequence diagrams. UML notation is accurate and consistently applied (lifelines, activation bars, messages, returns, conditions, loops). Interactions clearly reflect system behavior and align perfectly with functional requirements and class diagrams. Descriptions are clear, technically precise, and demonstrate deep understanding of object interactions and system flow.	logical message flow. Minor omissions or small notation inconsistencies may exist. Descriptions are clear but may lack deeper explanation of interaction logic. Good alignment with requirements and design.	lack some interaction details. Minor UML errors or missing elements (e.g., return messages, conditions). Descriptions are basic and mostly descriptive rather than analytical. Partial alignment with requirements.	notation contains noticeable errors. Interaction flow may be unclear or logically inconsistent. Descriptions are vague or insufficient. Weak alignment with system requirements.	UML errors present. Interaction flow does not reflect system behavior. No meaningful description provided.	
UML Class Diagram Design and Description	**10 to >7.92 pts Excellent** Class diagram is complete and technically accurate. All relevant classes, attributes, methods, visibility, relationships (association, aggregation, composition, inheritance), and multiplicities are correctly modeled using proper UML notation. Design clearly	**7.92 to >6.92 pts Good** Class diagram includes most required classes and relationships with generally correct UML notation. Minor omissions (e.g., missing multiplicity or visibility) or small inconsistencies may exist. Description explains structure adequately with good understanding of	**6.92 to >5.92 pts Satisfactory** Basic class diagram provided with key classes identified. Some relationships, attributes, or methods may be missing or simplified. Minor UML notation errors present. Description is mostly descriptive and lacks deeper design justification.	**5.92 to >4.92 pts Fair** Class diagram is partially complete with significant missing elements (e.g., relationships, methods, multiplicities). Noticeable UML errors or conceptual misunderstandings. Description is vague or insufficient. Weak alignment with requirements or sequence diagrams.	**4.92 to >0 pts Poor** Class diagram is missing or fundamentally incorrect. Major UML notation errors. Structure does not reflect system requirements. Little or no meaningful description provided.	10 pts

Criteria	Ratings					Pts
	reflects system requirements and aligns consistently with sequence diagrams and use cases. Accompanied by a clear, precise explanation demonstrating strong understanding of object-oriented principles (encapsulation, abstraction, modularity).	object-oriented design.				
Methods and Management	**5 to >3.95 pts Excellent** Project timelines and effort estimates are comprehensive, highly realistic, and clearly justified using the chosen software engineering methodology. Estimation techniques are well explained, consistently applied, and strongly aligned with project scope, resources, and deliverables.	**3.95 to >3.45 pts Good** Project timelines and estimates are clear and realistic, with good alignment to the selected methodology. Estimation approach is appropriate and mostly justified, with minor gaps in detail or consistency.	**3.45 to >2.95 pts Satisfactory** Project timelines and estimates are provided and generally aligned with the methodology, but justification or consistency of the estimation approach is limited. Some timelines may lack accuracy or detail.	**2.95 to >2.45 pts Fair** Project timelines and estimates are present but weakly connected to the chosen methodology. Estimation approach is unclear, incomplete, or inconsistently applied, resulting in questionable timelines.	**2.45 to >0 pts Poort** Project timelines and estimates are missing, unclear, or not based on any identifiable software engineering methodology. No meaningful justification or alignment with project scope or management approach.	5 pts
						Total Points: 50